

Network Science Assignment 1

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You can find the source code that generated the plots on the GitHub repository: [GitHub Repository](#), as well appended in this document. We both have equally worked on all parts of the assignment.

1 Permutations and Irreducibility

1.1 Taks 1

1.1.1 Vector multiplication

To show that the multiplication of matrix $P = P^{(i,j)}$ by a vector v has the effect of switching rows i and j of vector v , we consider:

$$P^{(i,j)} = \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix}, v = \begin{pmatrix} v_1 \\ \vdots \\ v_i \\ \vdots \\ v_j \\ \vdots \\ v_n \end{pmatrix}$$

We can see that the permutation matrix is an identity matrix with switch rows i and j . The multiplication of matrix $P = P^{(i,j)}$ and vector v is as

follows:

$$\begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_i \\ \vdots \\ v_j \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_j \\ \vdots \\ v_i \\ \vdots \\ v_n \end{pmatrix}$$

We can see that this permutation matrix switched the rows of i and j in the vector.

1.1.2 Left multiplication

To show that a left multiplication of matrix P by a $n \times m$ matrix A has the effect of switching rows i and j of matrix A, we consider:

$$P^{(i,j)} = \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix}, A = \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1j} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} & \cdots & a_{ij} & \cdots & a_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j1} & \cdots & a_{ji} & \cdots & a_{jj} & \cdots & a_{jm} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nj} & \cdots & a_{nm} \end{pmatrix}$$

The left multiplication of matrix $P = P^{(i,j)}$ and matrix A is as follows:

$$\begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1j} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} & \cdots & a_{ij} & \cdots & a_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j1} & \cdots & a_{ji} & \cdots & a_{jj} & \cdots & a_{jm} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nj} & \cdots & a_{nm} \end{pmatrix} =$$

$$\begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1j} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j1} & \cdots & a_{ji} & \cdots & a_{jj} & \cdots & a_{jm} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} & \cdots & a_{ij} & \cdots & a_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nj} & \cdots & a_{nm} \end{pmatrix}$$

We can see that the rows i and j switched with left multiplication.

1.1.3 Right multiplication

To show that a right multiplication of matrix P by a $m \times n$ matrix B has the effect of switching columns i and j of matrix A, we consider:

$$P^{(i,j)} = \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix}, B = \begin{pmatrix} b_{11} & \cdots & b_{1i} & \cdots & b_{1j} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{i1} & \cdots & b_{ii} & \cdots & b_{ij} & \cdots & b_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{j1} & \cdots & b_{ji} & \cdots & b_{jj} & \cdots & b_{jn} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mi} & \cdots & b_{mj} & \cdots & b_{mn} \end{pmatrix}$$

The left multiplication of matrix $P = P^{(i,j)}$ and matrix A is as follows:

$$\begin{pmatrix} b_{11} & \cdots & b_{1i} & \cdots & b_{1j} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{i1} & \cdots & b_{ii} & \cdots & b_{ij} & \cdots & b_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{j1} & \cdots & b_{ji} & \cdots & b_{jj} & \cdots & b_{jn} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mi} & \cdots & b_{mj} & \cdots & b_{mn} \end{pmatrix} \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix} =$$

$$\begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1i} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{i1} & \cdots & b_{ij} & \cdots & b_{ii} & \cdots & b_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{j1} & \cdots & b_{jj} & \cdots & b_{ji} & \cdots & b_{jn} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mj} & \cdots & b_{mi} & \cdots & b_{mn} \end{pmatrix}$$

We can see that the columns i and j switched with right multiplication.

1.2 Task 2

To prove that for any permutation matrix P, we have $P^T = P^{-1}$, we first have to state that every elementary permutation matrix is only one row swap from the identity matrix I.

$$I = \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix}, P^{(i,j)} = \begin{pmatrix} 1_1 & \cdots & 0_1 & \cdots & 0_1 & \cdots & 0_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_j & \cdots & 0_j & \cdots & 1_j & \cdots & 0_j \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_i & \cdots & 1_i & \cdots & 0_i & \cdots & 0_i \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0_n & \cdots & 0_n & \cdots & 0_n & \cdots & 1_n \end{pmatrix}$$

If we take the transpose of such a permutation matrix, it would lead to exactly the same matrix because there were only row swaps.

This is the same for the inverse because if we build an inverse of such en

elementary permutation matrix it will result the same matrix. The reason for that is when we build a matrix we firstly consider row swapping (Example below: Inverse is build with only swapping first and last row):

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array} \right]$$

This states that for elementary permutation matrices $P^T = P^{-1}$.

To prove that this also holds to every permutation matrix, we must consider that each permutation matrix (P_G) can be established by the product of multiple elementary permutation matrices ($P_1, P_2, P_3, \dots, P_K$).

$$P_G = P_1 * P_2 * \dots * P_k$$

$$(AB)^T = B^T * A^T$$

$$(AB)^{-1} = B^{-1} * A^{-1}$$

So we started some simple mathematical principles and now we can continue with the proof:

$$P_G^T = (P_1 * P_2 * \dots * P_k)^T = P_k^T * \dots * P_2^T * P_1^T$$

$$P_G^{-1} = (P_1 * P_2 * \dots * P_k)^{-1} = P_k^{-1} * \dots * P_2^{-1} * P_1^{-1}$$

$$P_G^T = P_k^T * \dots * P_2^T * P_1^T = P_k^{-1} * \dots * P_2^{-1} * P_1^{-1} = P_G^{-1}$$

$$P_G^T = P_G^{-1}$$

The third step holds since we proved before that for every elementary permutation matrices $P^T = P^{-1}$. Therefore we proved now that $P^T = P^{-1}$ also holds for all other permutation matrices.

1.3 Task 3

To show that the adjacency matrix A is reducible we have to find a permutation matrix P which makes the definition true. We know from task 1 that a left multiplication changes rows and a right multiplication changes columns. So to find out if such a P exists, we must find a way that P changes the same

rows and columns in A because we can consider $P^TAP=A$ because A fulfills the requirements of the solution.

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, A_{(1,1)} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, A_{(2,2)} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We see if we change for example rows 1,3 and 4,5 with left multiplication and after that columns 1,3 and 4,5 with right multiplication, it results in the same:

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

So that shows that the corresponding adjacency matrix is reducible because such the permutation matrix exists.

1.4 Task 4

Definition 3 states that an adjacency matrix is irreducible for connected graphs. Theorem 1, on the other hand, establishes a bi-directional relationship: a graph is connected if and only if its adjacency matrix is irreducible. While these mathematical concepts are valuable for theoretical analysis, there are practical challenges when it comes to real-world applications, especially for large networks. Here are some reasons why they may not be practical for verifying graph connectivity in real-world examples:

- **Matrix Size and Complexity:** In real-world applications, networks can be extremely large and complex. The adjacency matrix of a large graph can be prohibitively large and difficult to handle computationally. Determining the irreducibility of a large matrix can be a computationally intensive task, making it impractical.

- **Computational Complexity:** Verifying the irreducibility of a matrix typically involves complex matrix operations, such as matrix factorizations or iterative methods. These methods can be computationally expensive and may not scale well to large networks.
- **Sparse Graphs:** Many real-world networks are sparse, meaning that there are relatively few connections compared to the total number of potential connections. In such cases, the adjacency matrix is often sparse, and methods for checking irreducibility may not be directly applicable.
- **Numerical Approximations:** In practice, numerical approximations or heuristics may be used to determine connectivity. These approximations can provide a practical way to estimate connectivity but may not guarantee the mathematical property of irreducibility.
- **Graph Dynamics:** Real-world networks are not static; they can evolve and change over time. The concept of connectivity is dynamic in such cases. Theorems based on static matrices may not capture the evolving nature of networks.
- **Network Incompleteness:** In many cases, the adjacency matrix may be incomplete or noisy, making it challenging to apply theoretical definitions directly.
- **Alternative Connectivity Measures:** Real-world network analysis often involves more complex measures of connectivity, such as measures of reachability or network flow. These measures provide richer information about network behavior but may not align perfectly with the concept of irreducibility.
- **Network Visualization:** In practice, network visualization and graph analysis tools are often used to gain insights into network connectivity. These tools may provide a more intuitive and practical way to understand connectivity.

In summary, while the concept of irreducibility in adjacency matrices is valuable in graph theory, practical considerations such as computational complexity, network size, and the evolving nature of real-world networks make it challenging to directly use this concept to determine connectivity in

large, complex networks. Researchers and practitioners often rely on a combination of techniques, heuristics, and domain-specific knowledge to assess network connectivity in real-world applications.

1.5 Task 5

Theorem 1 says that a graph is only connected iff its adjacency matrix A is irreducible. Theorem 2 says that matrix A is irreducible iff $\forall i, j \in [1, \dots, n], \exists k = k(i, j) > 0$ such that $(A^k)_{ij} > 0$. So together this leads to a graph which is only connected iff $\forall i, j \in [1, \dots, n], \exists k = k(i, j) > 0$ such that $(A^k)_{ij} > 0$.

Since the graph is connected, we can conclude that for any pair of nodes i and j , there exists a path of length $k(i, j)$ between them. Therefore, $(A^k)_{ij}$ must be greater than 0 for every pair of nodes i and j , as required.

2 Clustering Coefficient

You can find our Coding Exercise with the complete Code in the Jupyter Notebook.

3 One-dimensional Lattice

3.1 Task 1

Definition 3 states that there is a chain C which is undirected and has n vertices. However, one of the most important part of this definition is this:

$$E = \{(i, i + 1), i = 1, \dots, n - 1\}$$

This statement makes it clear that the vertices are connected sequentially. This concludes that the distance is simply the number of vertices you need to pass to get to the target. Because the graph is undirected we have to consider two different cases:

1. When $i \leq j$ then it would go to the **right** direction. This would mean that the distance $d_{i,j} = j - i$
2. When $j \leq i$ then it would go to the **left** direction. This would mean that the distance $d_{i,j} = i - j$

We can combine both cases with the absolute value of the distance. This leads to the following result:

$$d_{i,j} = |i - j| \forall i, j \in \{1, \dots, n\}$$

3.2 Task 2

At first, we consider the symmetry that makes $d_{i,j} = d_{j,i}$ since there are the same number of distances that go right as distances go left. Also, i should never be the same as j because the distance between the same vertex is always zero and therefore it has no relevance in the summation. To summarize the following statement holds for symmetry reasons:

$$\sum_{i=1}^n \sum_{j=1}^n d_{i,j} = 2 * \sum_{i=1}^n \sum_{j=i+1}^n d_{i,j}$$

With this, the calculation looks like this now:

$$\langle d_{i,j} \rangle = 2 * \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=i+1}^n (j - i)$$

Now we calculate or simplify the inner summation first:

1. Split the inner summation in two separate:

$$\sum_{j=i+1}^n (j - i) = \sum_{j=i+1}^n j - \sum_{j=i+1}^n i$$

2. Simplify the first summation:

$$\sum_{j=i+1}^n j = \sum_{j=1}^n j - \sum_{j=1}^i j = \frac{n(n+1)}{2} - \frac{i(i+1)}{2}$$

3. Simplify the second summation:

$$\sum_{j=i+1}^n i = ni - i^2$$

So together the calculation looks now like this:

$$\langle d_{i,j} \rangle = 2 * \frac{1}{n(n-1)} \sum_{i=1}^n \left(\frac{n(n+1)}{2} - \frac{i(i+1)}{2} - i(n-i) \right)$$

So now we make some steps to simplify the summation:

1. Adjust the calculation:

$$\sum_{i=1}^n \left(\frac{n(n+1)}{2} - \frac{i(i+1)}{2} - i(n-i) \right) = \frac{n^2(n+1)}{2} - \frac{1}{2} \sum_{i=1}^n i(i+1) - \sum_{i=1}^n i(n-i)$$

2. Simplify the first summation:

$$\begin{aligned} \sum_{i=1}^n i(i+1) &= \sum_{i=1}^n i^2 + \sum_{i=1}^n i = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\ &= \frac{n(n+1)(2n+1+3)}{6} = \frac{2n(n+1)(n+2)}{6} = \frac{n(n+1)(n+2)}{3} \end{aligned}$$

3. Simplify the second summation:

$$\begin{aligned} \sum_{i=1}^n i(n-i) &= \sum_{i=1}^n in - \sum_{i=1}^n i^2 = \frac{n^2(n+1)}{2} - \frac{n(n+1)(2n+1)}{6} \\ &= \frac{n(n+1)(3n - (2n+1))}{6} = \frac{n(n+1)(n-1)}{6} = \frac{n(n^2-1)}{6} \end{aligned}$$

4. Gather everything together:

$$\sum_{i=1}^n \left(\frac{n(n+1)}{2} - \frac{i(i+1)}{2} - i(n-i) \right) = \frac{n^2(n+1)}{2} - \frac{n(n+1)(n+2)}{6} - \frac{n(n^2-1)}{6}$$

So at the end we only have to simplify everything:

$$\langle d_{i,j} \rangle = 2 * \frac{1}{n(n-1)} \frac{3n^2(n+1) - n(n+1)(n+2) - n(n^2-1)}{6}$$

$$\begin{aligned}
&= \frac{1}{n(n-1)} \frac{n(n+1)(n-1)}{3} = \frac{1}{n(n-1)} \frac{n(n^2-1)}{3} \\
&= \frac{n^2-1}{3(n-1)} = \frac{(n-1) * (n+1)}{3(n-1)} = \frac{n+1}{3}
\end{aligned}$$

So the explicit expression of the Average Distance of a chain C of length n is $\frac{n+1}{3}$.

3.3 Task 3

$$\begin{aligned}
\sum_j d_{i,j} &= \sum_{k=1}^{\lfloor n/2 \rfloor} k + \sum_{k=1}^{\lfloor n/2 \rfloor} k = \\
2 * \sum_{k=1}^{\lfloor n/2 \rfloor} k &= 2 * \frac{\lfloor n/2 \rfloor * (\lfloor n/2 \rfloor + 1)}{2} = \lfloor n/2 \rfloor * (\lfloor n/2 \rfloor + 1) \\
L_i &= \frac{1}{n-1} * \sum_j d_{i,j} = \frac{1}{n-1} * \lfloor n/2 \rfloor * (\lfloor n/2 \rfloor + 1) \\
&= \frac{1}{n-1} * \frac{n-1}{2} * \left(\frac{n-1}{2} + 1\right) = \frac{1}{2} * \left(\frac{n-1}{2} + 1\right) = \frac{n-1}{4} + \frac{1}{2} = \frac{n+1}{4}
\end{aligned}$$

3.4 Task 4

One of the most important info in this exercise is that n must be odd and $\frac{n-1}{2k}$ must be an integer. This has the effect, that there are always the same amount of distances in each 1-d lattice. Lets look at the adjacency matrix of some examples, where $A^{(n,k)}$:

$$A^{(5,2)} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}, A^{(9,2)} = \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 2 & 1 & 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 2 & 2 & 1 & 1 & 0 & 1 & 1 & 2 & 2 \\ 2 & 2 & 2 & 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 2 & 2 & 2 & 1 & 1 & 0 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 & 1 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}$$

We can see that there is always the same amount of distances. Also in every row there are the same numbers. So if we would take the sum of each row it would be the same result in every row. For that reason we only have to calculate the sum of one row and then divide it through the number of nodes (or number of nodes - 1 because 0 does not play a role).

To get to a formula we must follow this steps:

1. Because n has always to be odd $n - 1$ must be even. So $n - 1$ must be divisible by $2k$ and when we put this together we can see which will be the max distance. So:

$$d_{max} = \frac{n-1}{2k}$$

When we begin with $z = 1$ then we know it must end with $z = \frac{n-1}{2k}$ to get all distances.

2. So now we now which distances there are. We now have to calculate each distance with $2k$ because there are always k neighbours on the right and left side.
3. Now we have the sum of distances in one node. After that we can calculate the Average Distance of this node when we multiply the sum with $\frac{1}{n-1}$.
4. It is important to note that this Average Distance of a node is always the same. Therefore we do not have to calculate it for every node. The Average distance of one node is also the average distance of the whole 1-d lattice

With this we come to the explicit expression of the Average Distance of such a generic 1-d lattice:

$$D_{avg} = \frac{1}{n-1} * \sum_{z=1}^{\frac{n-1}{2k}} 2k * z$$

3.5 Task 5

Like in Task 4 we can see if we write down the adjacency matrix that each row has the same sum. So this means that the Average distance is the same for each row. So we only have to calculate the average distance of one row. This means we take the sum of all numbers in a row a then divide it with

(n-1).

The problem of Task 5 is that we do not have a rule like in Task 4 that $\frac{n-1}{2k}$ should be an integer. This means for example that when even only one node has a distance of 3 it should also be possible to calculate the Average Distance. Lets consider for example a 1-d lattice with coordination number $k = 2$ and number of nodes $n = 10$ then $\frac{n-1}{2k} = 2.25$ which is not an integer. This example has one single node which has a distance of 3 and all other has even distance of 2 or 1. To consider this there are multiple steps to take into account:

1. Firstly, we have to see if there is an integer when we calculate $\frac{n-1}{2k}$. When there is an integer then we know we have a symmetric row and we could take the formula of Task 4. However we have to find a way to write a formula that holds for everything. For this reason we can take the modulo of the rounded up and rounded down (see below). The reason is if there would be an integer it would result to zero and if not it would result to one. When multiplying with a term it is like a mathematical if condition because we would not take into account if zero and if one we would:

$$\lceil \frac{n-1}{2k} \rceil \bmod \lfloor \frac{n-1}{2k} \rfloor$$

2. Secondly, we have to know (if the modulo is not zero) how many times the max distance ($\lceil \frac{n-1}{2k} \rceil$) appear, to add this to the addition. This is done with this part:

$$((n-1) - \lfloor \frac{n-1}{2k} \rfloor * 2k)$$

This part will be multiplied by the max distance.

3. Lastly, at the end everything (the sum) must be divided with $(n-1)$ to get the average distance in a row. Remember from before, we only must calculate one row because in every row it will be the same.

The end formula looks like this:

$$\frac{1}{n-1} * (\lceil \frac{n-1}{2k} \rceil \bmod \lfloor \frac{n-1}{2k} \rfloor * (((n-1) - \lfloor \frac{n-1}{2k} \rfloor * 2k) * \lceil \frac{n-1}{2k} \rceil) + \sum_{z=1}^{\lfloor \frac{n-1}{2k} \rfloor} z * 2k)$$

The result of this formula will be the Average Distance for a 1-d lattice in the generic case where the condition of the exercise sheet holds.

4 Centralities

You can find our Coding Exercise with the complete Code in the Jupyter Notebook.

5 Code

Assignment_01

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1 Authors

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1.1 II - Clustering Coefficient - Coding Exercise (6 points)

For the provided datasets (you can find the dataset files in the Materials/Session 03 folder on OLAT):

1. Plot the average degree of the nearest neighbours knn as a function of the vertices degree.
2. Super impose the same quantity for the randomised network. Hint: The randomised networks are obtained via multiple edge swaps via the networkx function `nx.algorithms.smallworld.random_reference`. Make sure to set the parameter `connectivity = False` to have faster execution. If execution is still slow, we provide randomised datasets.)
3. Compute the assortativity coefficient of the real network.
4. Compute the assortativity coefficient of the randomised network.

```
[208]: # Imports
import numpy as np
import matplotlib.pyplot as plt
import networkx as nx
import os
import glob
import re
from tqdm import tqdm
from scipy.stats import pearsonr, spearmanr, kendalltau
import shutil
```

```
[209]: # Define colors with reduced opacity from a colormap
COLOR_BLUE = 'dodgerblue'
COLOR_RED = 'lightcoral'

# Define max number of k-neighbors
MAX_K = range(1, 10, 1)
RANGE_K = [1, 2, 5, 10, 20, 50, 100, 200]

KNN_RANGE = RANGE_K
```

```
# Define the main data directory
OUTPUT_PATH = "./output"
```

```
[210]: # Prepare output folder

# Delete the entire 'output' directory and its contents
if os.path.exists(OUTPUT_PATH):
    shutil.rmtree(OUTPUT_PATH)

# Create the 'output' directory
os.makedirs(OUTPUT_PATH)

# Define subdirectories
OUTPUT_PATH_AVG_DEGREE = os.path.join(OUTPUT_PATH, "avg_degree")
OUTPUT_PATH_RANDOMIZED = os.path.join(OUTPUT_PATH, "avg_degree_randomized")
OUTPUT_PATH_CENTRALITIES = os.path.join(OUTPUT_PATH, "centralities")

# Recreate the subdirectories
os.makedirs(OUTPUT_PATH_AVG_DEGREE)
os.makedirs(OUTPUT_PATH_RANDOMIZED)
os.makedirs(OUTPUT_PATH_CENTRALITIES)
```

1.1.1.1 Plot the average degree of the nearest neighbours knn as a function of the vertices degree.

```
[211]: # Define the directory containing the .gml files
directory = 'assignment_1_data'

# Use glob to get a list of all .gml files in the directory
gml_files = glob.glob(os.path.join(directory, '*.gml'))

# Init an empty dictionary for all graphs - the key is the actual name, and the
# value is the graph data
graphs = {}

# Use tqdm to show a progress bar in Jupyter Notebook
with tqdm(total=len(gml_files), desc="Loading GML Files") as pbar:
    # Loop through the list of .gml files and load them
    for file_path in gml_files:
        with open(file_path, 'r') as file:
            content = file.read()

            # Use regex to extract the desired part
            match = re.search(r'/(^[^/]+)\.gml$', file_path)
            if match:
                graph_name = match.group(1)
```

```

        # Add to the graph dictionary
        graph = nx.read_gml(file_path)
        graphs[graph_name] = graph

        # Print the graph name when each part is finished
        print(f"Finished loading {graph_name}")

    pbar.update(1) # Update the progress bar

```

Loading GML Files: 22%| | 2/9 [00:19<00:57, 8.21s/it]

Finished loading graph_AstroPh

Finished loading graph_jazz_collab

Loading GML Files: 33%| | 3/9 [00:26<00:45, 7.54s/it]

Finished loading graph_internet

Finished loading graph_game_thrones

Loading GML Files: 56%| | 5/9 [00:26<00:13, 3.33s/it]

Finished loading graph_eu_airlines

Loading GML Files: 67%| | 6/9 [00:35<00:14, 4.88s/it]

Finished loading graph_celegansInteractomes

Loading GML Files: 78%| | 7/9 [00:39<00:09, 4.68s/it]

Finished loading graph_facebook

Loading GML Files: 89%| | 8/9 [00:46<00:05, 5.17s/it]

Finished loading graph_CondMat

Loading GML Files: 100%| | 9/9 [00:49<00:00, 5.46s/it]

Finished loading graph_chess

```

[212]: # Initialize results dictionary
       results = {graph_name: {} for graph_name in graphs.keys()}

```

```

[213]: print(results)

```

```

{'graph_AstroPh': {}, 'graph_jazz_collab': {}, 'graph_internet': {},
 'graph_game_thrones': {}, 'graph_eu_airlines': {}, 'graph_celegansInteractomes':
 {}, 'graph_facebook': {}, 'graph_CondMat': {}, 'graph_chess': {}}

```

```

[214]: num_graphs = len(graphs)

```

```

[215]: DATASETS_A2 = ["graph_celegansInteractomes", "graph_AstroPh", "graph_CondMat",
                    ↵ "graph_chess", "graph_facebook", "graph_internet"]

```

```

[216]: def create_main_average_knn_plot(graphs, output_path, color, randomized=False):
    # Create a grid of subplots with 2 rows and 3 columns
    fig, axs = plt.subplots(2, 3, figsize=(20, 15))
    plt.subplots_adjust(wspace=0.2, hspace=0.2)

    legend_labels = [] # Create an empty list to store legend labels

    for j, (graph_name, graph) in enumerate(graphs.items()):
        if graph_name in DATASETS_A2:

            # Create a new figure and axis for each subplot
            row = j // 3 # Calculate the row index (0 or 1)
            col = j % 3 # Calculate the column index (0, 1, or 2)
            ax = axs[row, col]

            average_degree_connectivity = nx.
            average_degree_connectivity(G=graph)

            average_knn_x_label = f"Average k-nearest neighbour Degree (x){' ' if randomized else ''}"
            average_knn_y_label = f"Average k-nearest neighbour Degree{' ' if randomized else ''}"

            # Append to result dict

            results[graph_name][average_knn_x_label] =
            list(average_degree_connectivity.keys())
            results[graph_name][average_knn_y_label] =
            list(average_degree_connectivity.values())

            # Plot on the current subplot
            scatter = ax.scatter(x=results[graph_name][f"Average k-nearest neighbour Degree (x){' ' if randomized else ''}"],
            y=results[graph_name][f"Average k-nearest neighbour Degree{' ' if randomized else ''}"], color=color)
            ax.set_xlabel("Vertex Degree")
            ax.set_ylabel(f"Average k nearest neighbor")
            ax.set_title(f"Average k nearest neighbor vs. Vertex Degree{' ' if randomized else ''}:\n {graph_name}")
            ax.grid(True)

            # Append handles for legends
            legend_labels.extend([scatter])
        else:
            continue

```

```

    # Set title
    fig.suptitle(t=f"Average k nearest neighbors as a function of the vertices_
degree[' (randomized)' if randomized else '']", fontsize=18)

    # Create a shared legend with custom labels and colors
    legend_labels = [scatter] # Customize as needed
    legend_color = [COLOR_RED] if randomized else [COLOR_BLUE] # Customize as
needed
    if randomized:
        fig.legend(legend_labels, labels=['Randomized'], loc='upper right',
title="Legend", labelcolor=legend_color)
    else:
        fig.legend(legend_labels, labels=['Original'], loc='upper right',
title="Legend", labelcolor=legend_color)

    # Save the complete plot as a PNG image
    complete_plot_filename = os.path.join(output_path,
f"average_k_nearest{'_randomized' if randomized else ''}.png")
    plt.savefig(complete_plot_filename)

    plt.show()

```

```

[217]: graphs_t1 = {graph_name: graph for graph_name, graph in graphs.items() if
graph_name in DATASETS_A2}

```

```

[218]: create_main_average_knn_plot(graphs_t1, OUTPUT_PATH_AVG_DEGREE, COLOR_BLUE,
False)

```

```

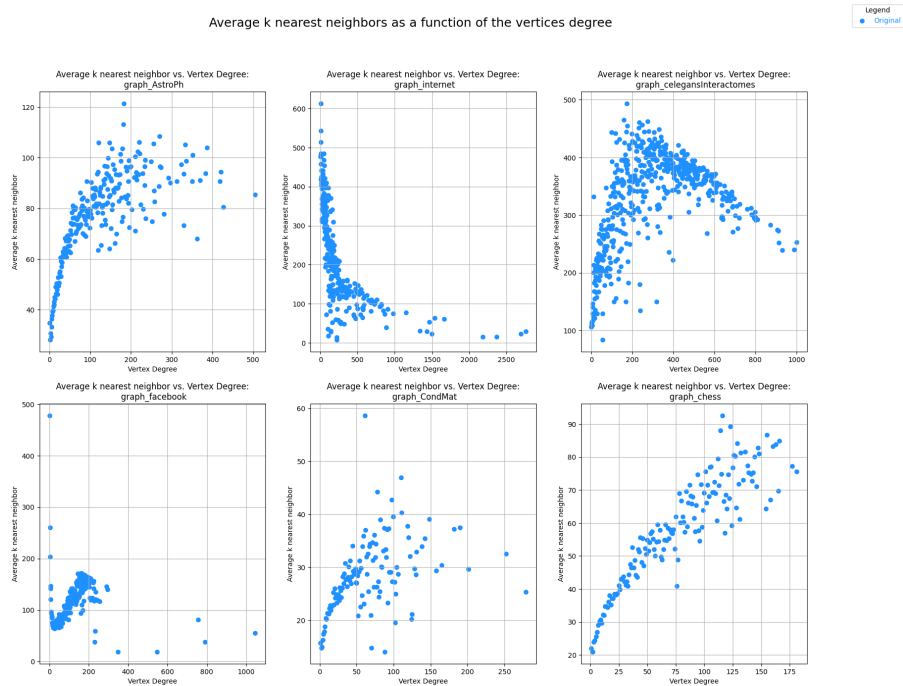
/var/folders/ck/vgxn_9f907d122j56f3203gm0000gn/T/ipykernel_62564/1678400520.py:4
8: UserWarning: You have mixed positional and keyword arguments, some input may
be discarded.

```

```

    fig.legend(legend_labels, labels=['Original'], loc='upper right',
title="Legend", labelcolor=legend_color)

```



1.2 2. Super impose the same quantity for the randomised network.

Hint: The randomised networks are obtained via multiple edge swaps via the networkx function `nx.algorithms.smallworld.random_reference`. Make sure to set the parameter `connectivity = False` to have faster execution. If execution is still slow, we provide randomised datasets.)

```
[219]: random_graphs = graphs.copy()
        SEED = 42

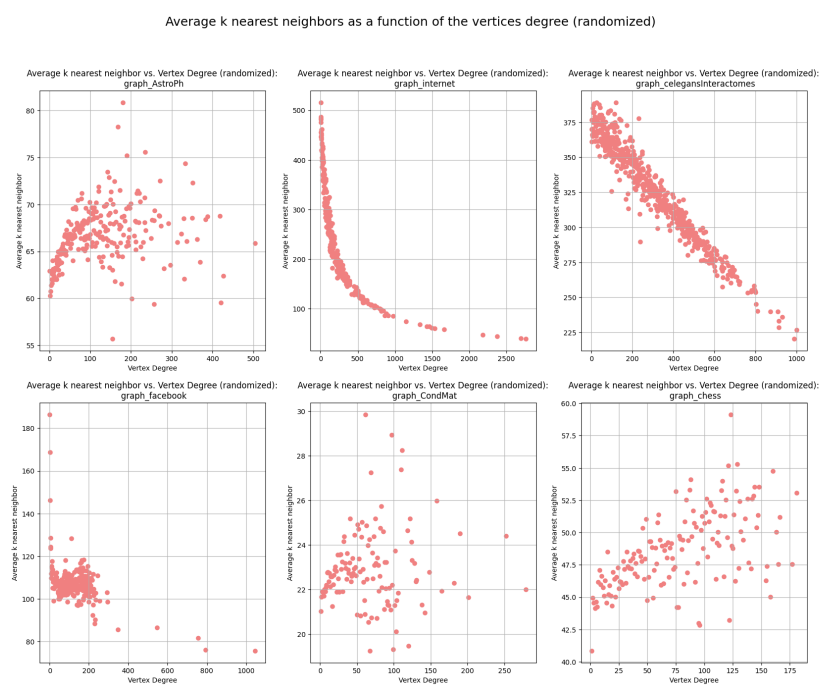
        for graph_name, g in random_graphs.items():
            random_graphs[graph_name] = nx.algorithms.smallworld.random_reference(G=g,
            niter=1, connectivity=False, seed=SEED)
```

```
[220]: graphs_randomized_t2 = {graph_name: graph for graph_name, graph in
    random_graphs.items() if graph_name in DATASETS_A2}
```

```
[221]: create_main_average_knn_plot(graphs_randomized_t2, OUTPUT_PATH_RANDOMIZED,
    COLOR_RED, True)
```

```
/var/folders/ck/vgxn_9f907d122j56f3203gm0000gn/T/ipykernel_62564/1678400520.py:4
6: UserWarning: You have mixed positional and keyword arguments, some input may
be discarded.
```

```
fig.legend(legend_labels, labels=['Randomized'], loc='upper right',
title="Legend", labelcolor=legend_color)
```



```
[222]: fig, axs = plt.subplots(2, 3, figsize=(20, 15))

plt.subplots_adjust(wspace=0.2, hspace=0.2)

legend_labels = [] # Create an empty list to store legend labels

for j, (graph_name, graph) in enumerate(graphs_randomized_t2.items()):
    if graph_name in DATASETS_A2:
        # Create a new figure and axis for each subplot
        row = j // 3 # Calculate the row index (0 or 1)
        col = j % 3 # Calculate the column index (0, 1, or 2)
        ax = axs[row, col]

        # Plot on the current subplot
```

```

        original_scatter = ax.scatter(x=results[graph_name]["Average k-nearest_
↵neighbour Degree (x)"], y=results[graph_name]["Average k-nearest neighbour_
↵Degree"], color=COLOR_BLUE)
        randomized_scatter = ax.scatter(x=results[graph_name]["Average_
↵k-nearest neighbour Degree (x) randomized"], y=results[graph_name]["Average_
↵k-nearest neighbour Degree randomized"], color=COLOR_RED)
        ax.set_xlabel("Vertex Degree")
        ax.set_ylabel("Average k nearest neighbor")
        ax.set_title(f"Super imposed average k nearest neighbor vs. Vertex_
↵Degree \n{graph_name}")
        ax.grid(True)

        # Append handles for legends
        legend_labels.extend([original_scatter, randomized_scatter])
    else:
        continue

# Set title
fig.suptitle(t=f"Average k nearest neighbors as a function of the vertices_
↵degree", fontsize=18)

# Create a shared legend with custom labels and colors
legend_labels = [original_scatter, randomized_scatter] # Customize as needed
legend_colors = [COLOR_BLUE, COLOR_RED] # Customize as needed
fig.legend(legend_labels, labels=['Original', 'Randomized'], loc='upper right',
↵title="Legend", labelcolor=legend_colors)

# Save the complete plot as a PNG image
complete_plot_randomized_filename = os.path.join(OUTPUT_PATH,
↵f"average_degree_for_knn_super_imposed.png")
plt.savefig(complete_plot_randomized_filename)

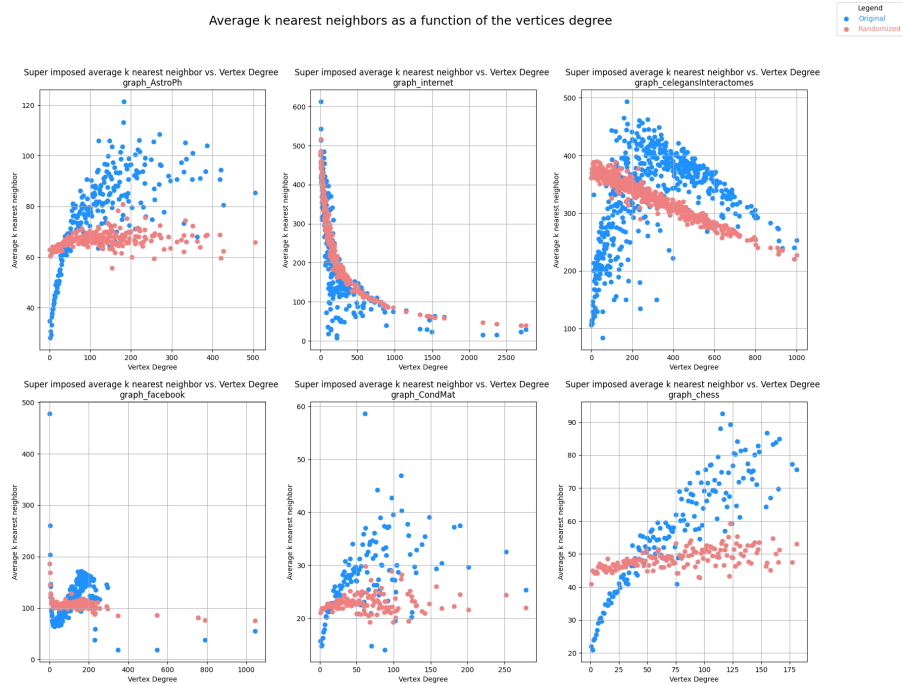
plt.show()

```

```

/var/folders/ck/vgxn_9f907d122j56f3203gm0000gn/T/ipykernel_62564/2456965222.py:3
3: UserWarning: You have mixed positional and keyword arguments, some input may
be discarded.
    fig.legend(legend_labels, labels=['Original', 'Randomized'], loc='upper
right', title="Legend", labelcolor=legend_colors)

```



1.3 3. Compute the assortativity coefficient of the real network.

```
[223]: def print_network_assortativity_coefficients(graphs, randomized=False):
# Calculate the maximum length of network names
max_name_length = max(len(graph_name) for graph_name in graphs.keys())

for graph_name, g in graphs.items():
    assortativity_coefficient = nx.degree_assortativity_coefficient(G=g)
    randomized_suffix = " (randomized)" if randomized else ""
    results[graph_name][f"Assortativity Coefficient{randomized_suffix}"] = assortativity_coefficient
    print(f"The network {graph_name:{max_name_length}s}{randomized_suffix} has an Assortativity Coefficient of {assortativity_coefficient:10.4f}")

[224]: print_network_assortativity_coefficients(graphs, False)
```

```
The network graph_AstroPh          has an Assortativity Coefficient of
0.2013
The network graph_jazz_collab       has an Assortativity Coefficient of
0.0202
```

The network graph_internet	has an Assortativity Coefficient of
-0.2149	
The network graph_game_thrones	has an Assortativity Coefficient of
-0.1308	
The network graph_eu_airlines	has an Assortativity Coefficient of
-0.1518	
The network graph_celegansInteractomes	has an Assortativity Coefficient of
0.1517	
The network graph_facebook	has an Assortativity Coefficient of
0.0636	
The network graph_CondMat	has an Assortativity Coefficient of
0.1253	
The network graph_chess	has an Assortativity Coefficient of
0.3705	

1.4 4. Compute the assortativity coefficient of the randomised network.

```
[225]: print_network_assortativity_coefficients(random_graphs, True)
```

The network graph_AstroPh	(randomized) has an Assortativity
Coefficient of 0.0196	
The network graph_jazz_collab	(randomized) has an Assortativity
Coefficient of -0.0906	
The network graph_internet	(randomized) has an Assortativity
Coefficient of -0.1976	
The network graph_game_thrones	(randomized) has an Assortativity
Coefficient of -0.1701	
The network graph_eu_airlines	(randomized) has an Assortativity
Coefficient of -0.2254	
The network graph_celegansInteractomes	(randomized) has an Assortativity
Coefficient of -0.1613	
The network graph_facebook	(randomized) has an Assortativity
Coefficient of -0.0319	
The network graph_CondMat	(randomized) has an Assortativity
Coefficient of 0.0111	
The network graph_chess	(randomized) has an Assortativity
Coefficient of 0.0428	

```
[226]: def print_network_comparison(results):
    # Determine the maximum length for network names
    max_name_length = max(len(graph_name) for graph_name in results.keys())

    print("Comparison of the assortativity coefficient in normal vs. randomized_
    networks")

    for graph_name, res in results.items():
        normal_coefficient = res.get('Assortativity Coefficient', 0.0)
```

```

        randomized_coefficient = res.get('Assortativity Coefficient_
        ↵(randomized)', 0.0)

        print(f"{graph_name:{max_name_length}s} - Normal: {normal_coefficient:
        ↵10.4f} Randomized: {randomized_coefficient:10.4f}")

```

```
[227]: print_network_comparison(results)
```

Comparison of the assortativity coefficient in normal vs. randomized networks

graph_AstroPh	- Normal:	0.2013	Randomized:	0.0196
graph_jazz_collab	- Normal:	0.0202	Randomized:	-0.0906
graph_internet	- Normal:	-0.2149	Randomized:	-0.1976
graph_game_thrones	- Normal:	-0.1308	Randomized:	-0.1701
graph_eu_airlines	- Normal:	-0.1518	Randomized:	-0.2254
graph_celegansInteractomes	- Normal:	0.1517	Randomized:	-0.1613
graph_facebook	- Normal:	0.0636	Randomized:	-0.0319
graph_CondMat	- Normal:	0.1253	Randomized:	0.0111
graph_chess	- Normal:	0.3705	Randomized:	0.0428

- The assortativity coefficient ranges from -1 to 1:
- A positive coefficient (closer to 1) indicates assortative mixing, meaning nodes with similar degrees tend to connect.
- A negative coefficient (closer to -1) indicates disassortative mixing, meaning nodes with dissimilar degrees tend to connect.
- A coefficient near 0 suggests a network with little to no assortative or disassortative mixing, indicating a more random or heterogeneous connection pattern.

1.5 IV - Centralities - Coding Exercise (9 points)

1. Compute degree, closeness, betweenness and eigenvector centrality for each node.
2. Do a scatter plot of each pair of centrality (6 plots total). Compute Pearson, Spearman and Kendall correlation coefficient for each pair and note them on the scatter plots. Hint: Centralities measures are available on NetworkX, while correlation coefficients are available on the module scipy.stats.
3. Repeat point 2 on randomised versions of the graphs provided. Hint: Check point 2 of Exercise 2, for indications on the randomisation function to use in NetworkX.
4. Briefly explain, for the Jazz collaboration network from Gleiser and Danon [2003], what each centrality measure means in practice.

```
[228]: # Define CENTRALITY_MEASURES
CENTRALITY_MEASURES = ["Degree", "Closeness", "Betweenness", "Eigenvector_
        ↵Centrality"]

```

```
[229]: SELECTED_GRAPHS = ["graph_jazz_collab", "graph_game_thrones",
        ↵"graph_eu_airlines"]

```

```
[230]: # create result dict for every graph
results_4 = dict.fromkeys(graphs)
for key, value in results_4.items():
    results_4[key] = {}
```

```
[231]: # find smallest graph
for name, g in graphs.items():
    print(len(g.nodes()), name)
```

```
17903 graph_AstroPh
198 graph_jazz_collab
34761 graph_internet
107 graph_game_thrones
417 graph_eu_airlines
5966 graph_celegansInteractomes
4039 graph_facebook
21363 graph_CondMat
7115 graph_chess
```

1.6 1. Compute degree, closeness, betweenness and eigenvector centrality for each node.

```
[232]: # Compute different metrics for all nodes in the graph
for i, (graph_name, g) in enumerate(graphs.items()):
    if graph_name in SELECTED_GRAPHS:
        node_properties = {}
        node_properties["Degree"] = nx.degree_centrality(G=g)
        node_properties["Closeness"] = nx.closeness_centrality(G=g)
        node_properties["Betweenness"] = nx.betweenness_centrality(G=g)
        node_properties["Eigenvector Centrality"] = nx.
        eigenvector_centrality(G=g)
        results_4[graph_name] = node_properties
```

```
[233]: # Remove graphs that are not part of the exercise (e.g., AstroPH, internet, ...)

# Get a list of keys with empty dictionaries
keys_to_remove = [key for key, value in results_4.items() if not bool(value)]

# Remove the keys from the dictionary
for key in keys_to_remove:
    del results_4[key]
```

- 1.7 2. Do a scatter plot of each pair of centrality (6 plots total). Compute Pearson, Spearman and Kendall correlation coefficient for each pair and note them on the scatter plots. Hint: Centralities measures are available on NetworkX, while correlation coefficients are available on the module `scipy.stats`.

```
[234]: def plot_centrality_scatter_and_correlation(result_dict, graph_list,
    randomized=False):

    if randomized:
        for g in range(len(graph_list)):
            graph_list[g] += "_randomized"

    # Create three separate plots for each network
    for graph_name, node_properties in result_dict.items():
        if graph_name in graph_list:
            # Create a grid of subplots with rows for each correlation measure
            # and columns for each centrality pair
            fig, axs = plt.subplots(len(CENTRALITY_MEASURES),
            len(CENTRALITY_MEASURES), figsize=(18, 18))

            # Adjust the spacing between subplots
            plt.subplots_adjust(wspace=0.4, hspace=0.4)

            # Iterate through centrality pairs
            for i in range(len(CENTRALITY_MEASURES)):
                for j in range(len(CENTRALITY_MEASURES)):
                    if i != j: # Avoid plotting the same centrality against
                    itself

                        x_centrality = CENTRALITY_MEASURES[i]
                        y_centrality = CENTRALITY_MEASURES[j]

                        x_data = list(node_properties[x_centrality].values())
                        y_data = list(node_properties[y_centrality].values())

                        # Determine subplot position
                        row = i
                        col = j

                        # Define color
                        color = COLOR_RED if randomized else COLOR_BLUE

                        # Create a scatter plot in the corresponding subplot
                        axs[row, col].scatter(x_data, y_data, color=color,
                        alpha=0.5)

                        axs[row, col].set_xlabel(f"{x_centrality}")
                        axs[row, col].set_ylabel(f"{y_centrality}")
```

```

        axes[row, col].set_title(f"{x_centrality} vs. {y_centrality}")

        # Calculate Pearson correlation coefficient
        pearson_corr, _ = pearsonr(x_data, y_data)

        # Calculate Spearman correlation coefficient
        spearman_corr, _ = spearmanr(x_data, y_data)

        # Calculate Kendall correlation coefficient
        kendall_corr, _ = kendalltau(x_data, y_data)

        # Note the correlation coefficients on the scatter plot
        corr_text = f"Pearson: {pearson_corr:.4f}\nSpearman: {spearman_corr:.4f}\nKendall: {kendall_corr:.4f}"

        # Place the correlation text below the plot
        axes[row, col].text(0.5, -0.3, corr_text, transform=axes[row, col].transAxes, fontsize=7, ha='center')

    else:
        # Skip plotting when the graph is not in the specified list
        axes[i, j].axis('off')

    # Adjust subplot layout
    plt.suptitle(f"Centrality Measures Scatter Plots and Correlation Coefficients - {graph_name}", fontsize=16)
    plt.tight_layout(rect=[0, 0, 1, 0.95]) # Adjust the supertitle position

    plt_filename = os.path.join(OUTPUT_PATH_CENTRALITIES, f"{graph_name}_centralities.png")
    plt.savefig(plt_filename)
    plt.show()

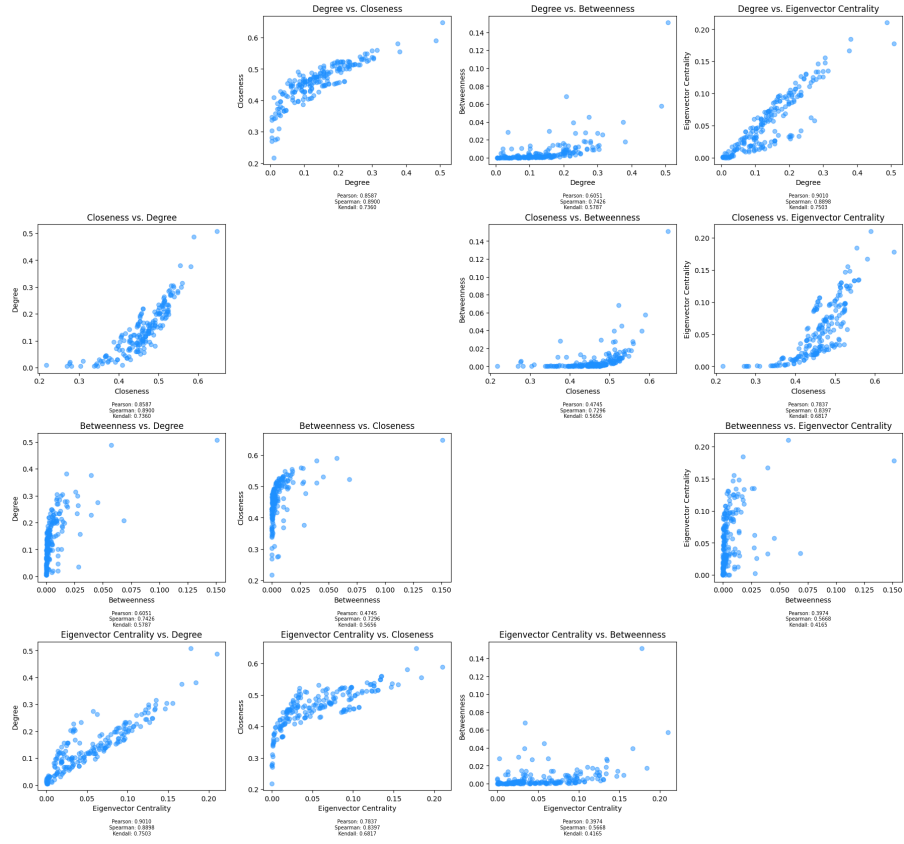
```

```

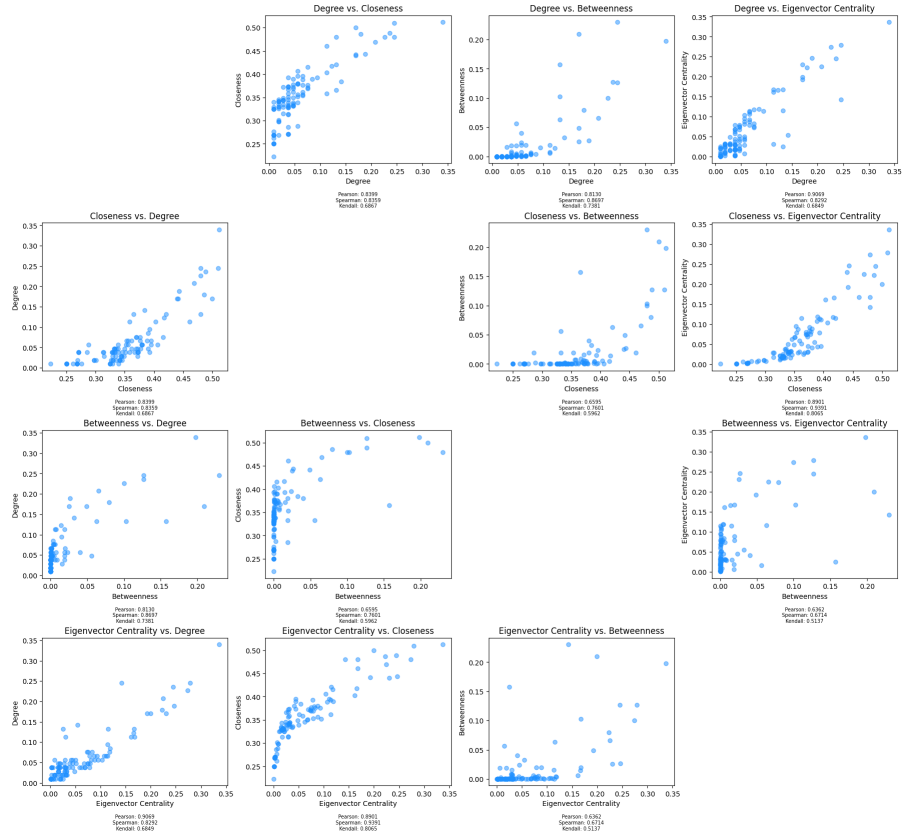
[235]: plot_centrality_scatter_and_correlation(result_dict=results_4, graph_list=SELECTED_GRAPHS, randomized=False)

```

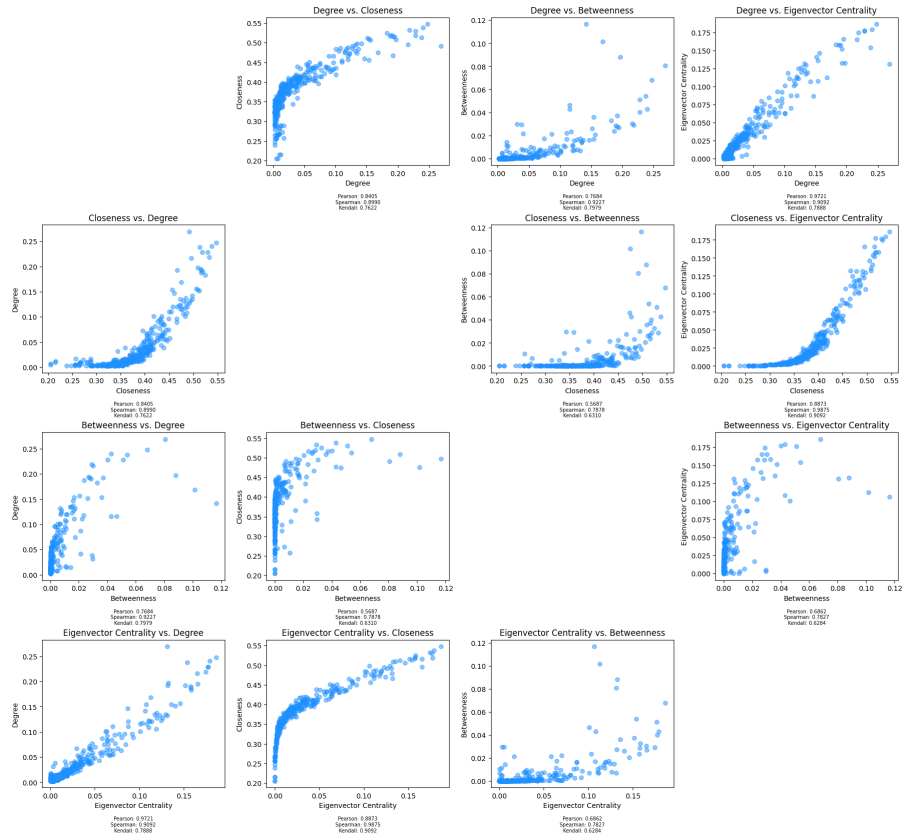
Centrality Measures Scatter Plots and Correlation Coefficients - graph_jazz_collab



Centrality Measures Scatter Plots and Correlation Coefficients - graph_game_thrones



Centrality Measures Scatter Plots and Correlation Coefficients - graph_eu_airlines



1.8 3. Repeat point 2 on randomised versions of the graphs provided. Hint: Check point 2 of Exercise 2, for indications on the randomisation function to use in NetworkX.

```
[236]: keys_to_add = []

# Iterate through the original keys
for key in results_4.keys():
    # Create a new key with the desired format and add it to the list
    keys_to_add.append(key + "_randomized")

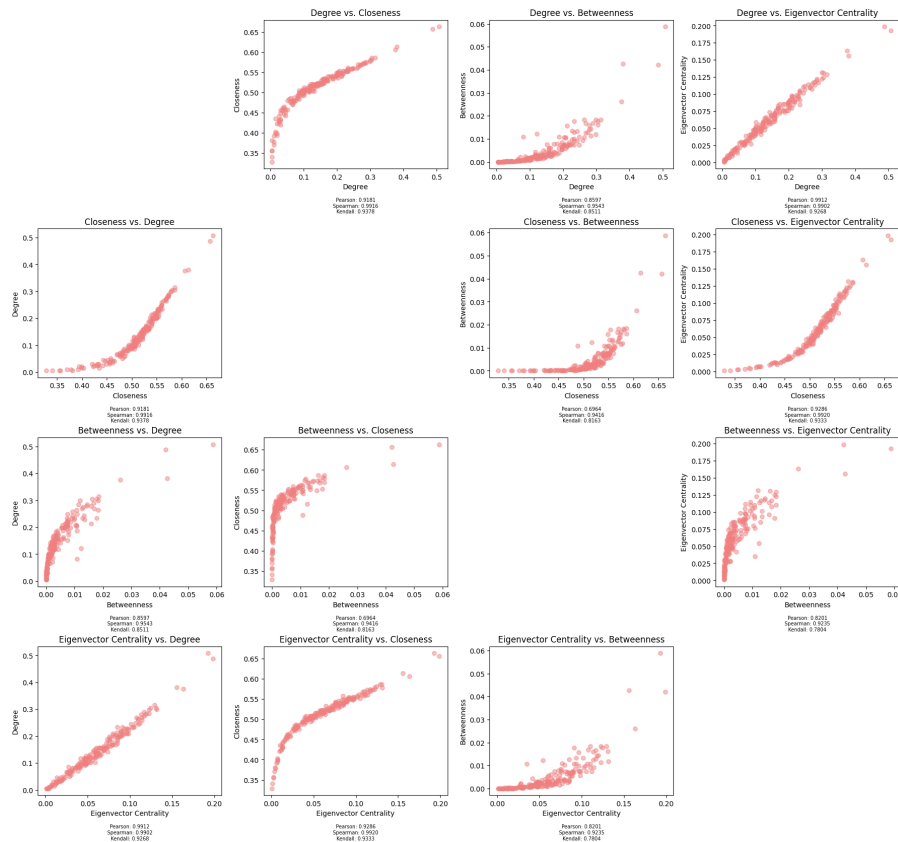
# Add the new keys to the dictionary
for new_key in keys_to_add:
```

```
results_4[new_key] = {}
```

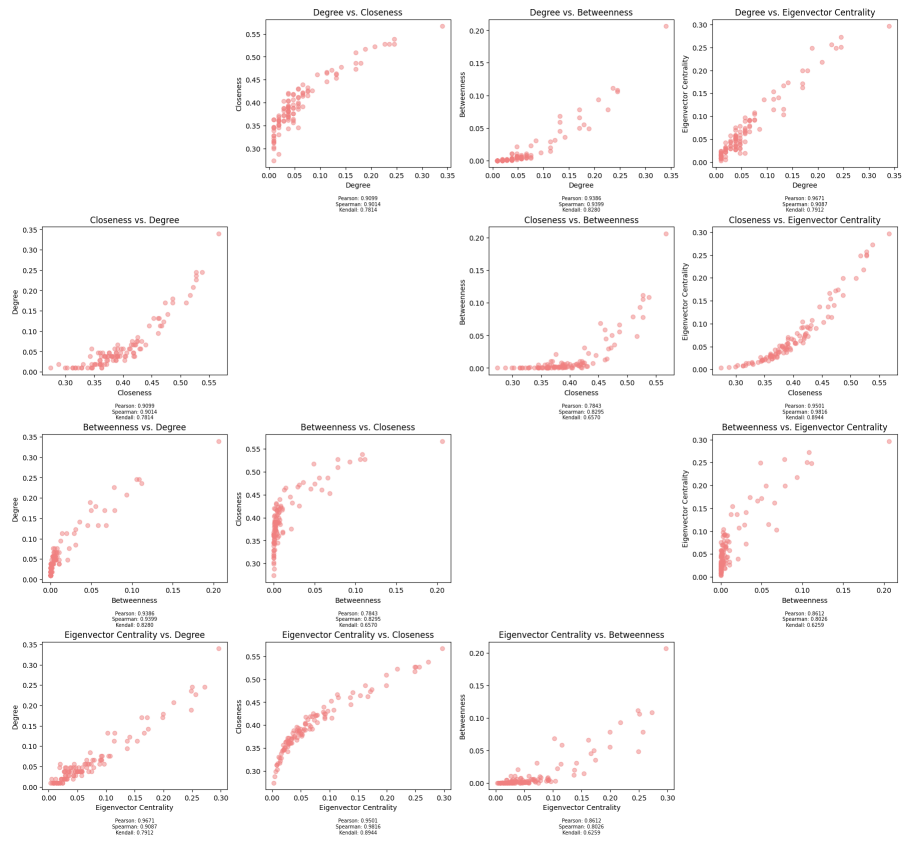
```
[237]: # Compute different metrics for all nodes in the graph
for i, (graph_name, g) in enumerate(random_graphs.items()):
    if graph_name in SELECTED_GRAPHS:
        node_properties = {}
        node_properties["Degree"] = nx.degree_centrality(G=g)
        node_properties["Closeness"] = nx.closeness_centrality(G=g)
        node_properties["Betweenness"] = nx.betweenness_centrality(G=g)
        node_properties["Eigenvector Centrality"] = nx.
        eigenvector_centrality(G=g)
        results_4[graph_name + "_randomized"] = node_properties
```

```
[238]: plot_centrality_scatter_and_correlation(result_dict=results_4,
        graph_list=SELECTED_GRAPHS, randomized=True)
```

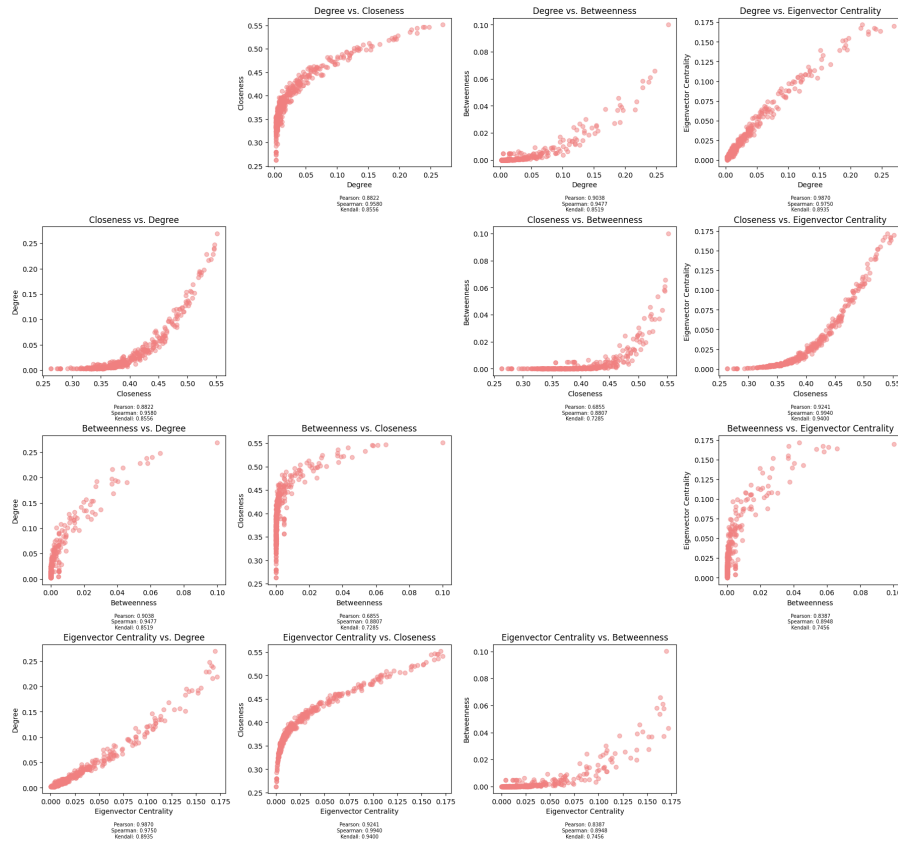
Centrality Measures Scatter Plots and Correlation Coefficients - graph_jazz_collab_randomized



Centrality Measures Scatter Plots and Correlation Coefficients - graph_game_thrones_randomized



Centrality Measures Scatter Plots and Correlation Coefficients - graph_eu_airlines_randomized



```
[239]: # Create three separate plots for each network
for graph_name, node_properties in results_4.items():
    if graph_name in ["graph_jazz_collab", "graph_game_thrones",
    ↪ "graph_eu_airlines"]:
        # Create a grid of subplots with rows for each correlation measure and
        ↪ columns for each centrality pair
        fig, axs = plt.subplots(len(CENTRALITY_MEASURES),
        ↪ len(CENTRALITY_MEASURES), figsize=(18, 18))

        # Adjust the spacing between subplots
        plt.subplots_adjust(wspace=0.4, hspace=0.4)

        # Iterate through centrality pairs
```

```

for i in range(len(CENTRALITY_MEASURES)):
    for j in range(len(CENTRALITY_MEASURES)):
        if i != j: # Avoid plotting the same centrality against itself
            x_centrality = CENTRALITY_MEASURES[i]
            y_centrality = CENTRALITY_MEASURES[j]

            # Original data in blue
            x_data = list(node_properties[x_centrality].values())
            y_data = list(node_properties[y_centrality].values())

            # Check if randomized data exists for the current graph
            randomized_graph_name = f"{graph_name}_randomized"
            if randomized_graph_name in results_4:
                randomized_node_properties = {}
                results_4[randomized_graph_name]
                randomized_x_data = {}
                list(randomized_node_properties[x_centrality].values())
                randomized_y_data = {}
                list(randomized_node_properties[y_centrality].values())
            else:
                randomized_x_data = []
                randomized_y_data = []

            # Determine subplot position
            row = i
            col = j

            # Create a scatter plot for original data (blue points)
            axs[row, col].scatter(x_data, y_data, color=COLOR_BLUE,
            alpha=0.5, label='Original Data')
            axs[row, col].set_xlabel(f"{x_centrality}")
            axs[row, col].set_ylabel(f"{y_centrality}")
            axs[row, col].set_title(f"{x_centrality} vs. {y_centrality}")

            # Create a scatter plot for randomized data (red points)
            axs[row, col].scatter(randomized_x_data, randomized_y_data,
            color=COLOR_RED, alpha=0.5, label='Randomized Data')

            # Calculate Pearson correlation coefficient for original
            data
            pearson_corr, _ = pearsonr(x_data, y_data)

            # Calculate Spearman correlation coefficient for original
            data
            spearman_corr, _ = spearmanr(x_data, y_data)

```

```

        # Calculate Kendall correlation coefficient for original
data    kendall_corr, _ = kendalltau(x_data, y_data)

        # Calculate Pearson correlation coefficient for randomized
data    randomized_pearson_corr, _ = pearsonr(randomized_x_data,
randomized_y_data)

        # Calculate Spearman correlation coefficient for randomized
data    randomized_spearman_corr, _ = spearmanr(randomized_x_data,
randomized_y_data)

        # Calculate Kendall correlation coefficient for randomized
data    randomized_kendall_corr, _ = kendalltau(randomized_x_data,
randomized_y_data)

        # Generate some descriptive text
        corr_text_original = f"Original Data:\nPearson:
{pearson_corr:.4f}\nSpearman: {spearman_corr:.4f}\nKendall: {kendall_corr:.
4f}"
        corr_text_randomized = f"Randomized Data:\nPearson:
{randomized_pearson_corr:.4f}\nSpearman: {randomized_spearman_corr:.
4f}\nKendall: {randomized_kendall_corr:.4f}"

        axs[row, col].text(0.1, -0.2, corr_text_original,
transform=axs[row, col].transAxes, fontsize=7, ha='center')
        axs[row, col].text(0.9, -0.2, corr_text_randomized,
transform=axs[row, col].transAxes, fontsize=7, ha='center')
        axs[row, col].legend(loc='upper left') # Add a legend

    else:
        # Skip plotting when the graph is not in the specified list
        axs[i, j].axis('off')

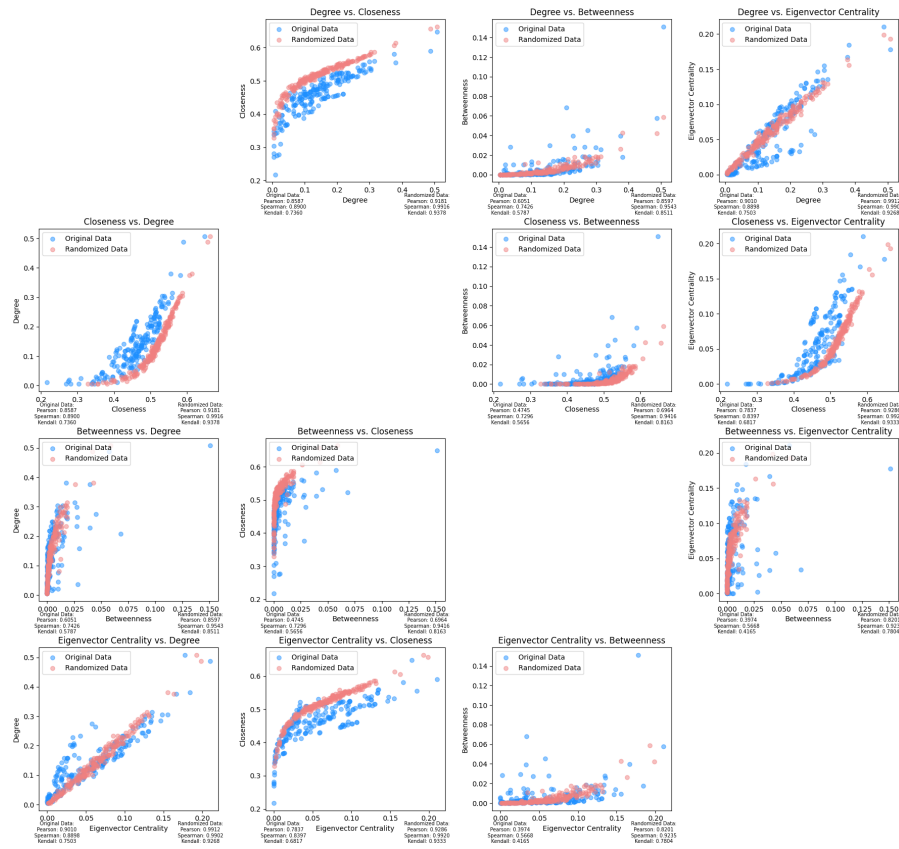
    # Adjust subplot layout
    plt.suptitle(f"Centrality Measures Scatter Plots and Correlation
Coefficients (normal and randomized) - {graph_name}", fontsize=16)
    plt.tight_layout(rect=[0, 0, 1, 0.95]) # Adjust the supitle position

    plt_filename = os.path.join(OUTPUT_PATH_CENTRALITIES,
f"{graph_name}_centralities_super_imposed.png")

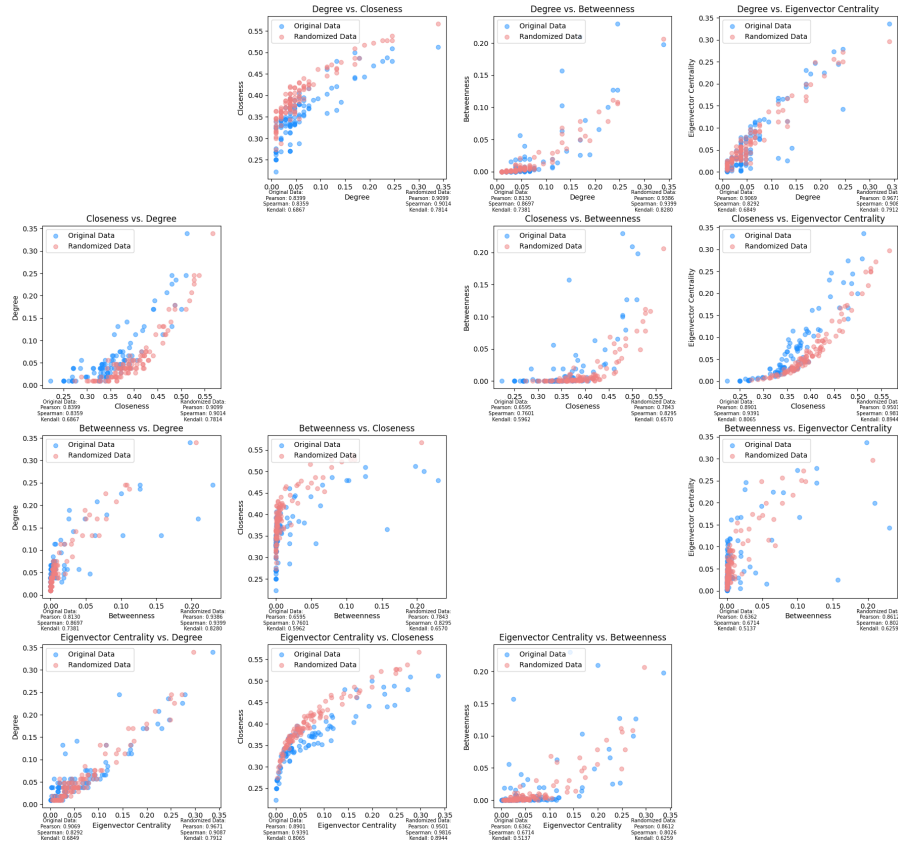
```

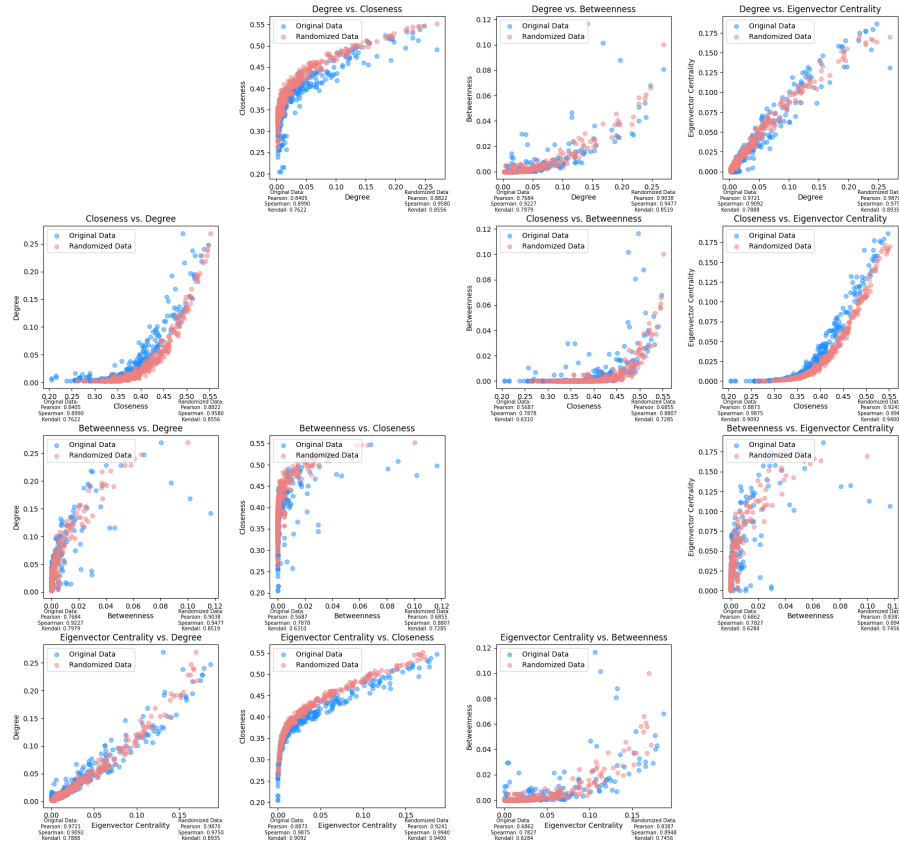
```
plt.savefig(plt_filename)
plt.show()
```

Centrality Measures Scatter Plots and Correlation Coefficients (normal and randomized) - graph_jazz_collab



Centrality Measures Scatter Plots and Correlation Coefficients (normal and randomized) - graph_game_thrones





1.9 4. Briefly explain, for the Jazz collaboration network from Gleiser and Danon [2003], what each centrality measure means in practice.

```
[240]: # Create three separate plots for each network
for graph_name, node_properties in results_4.items():
    if graph_name in ["graph_jazz_collab"]:
        # Create a grid of subplots with 2 rows and 3 columns
        fig, axs = plt.subplots(2, 3, figsize=(18, 12))

        # Adjust the spacing between subplots
        plt.subplots_adjust(wspace=0.4, hspace=0.4)
```

```

# Initialize row and col indices
row = 0
col = 0

# Iterate through centrality pairs
for i in range(len(CENTRALITY_MEASURES)):
    for j in range(i+1, len(CENTRALITY_MEASURES)): # Start from i+1 to
↳avoid duplicates
        if row < 2 and col < 3: # Stay within the grid
            x_centrality = CENTRALITY_MEASURES[i]
            y_centrality = CENTRALITY_MEASURES[j]

            # Original data in blue
            x_data = list(node_properties[x_centrality].values())
            y_data = list(node_properties[y_centrality].values())

            # Check if randomized data exists for the current graph
            randomized_graph_name = f"{graph_name}_randomized"
            if randomized_graph_name in results_4:
                randomized_node_properties =
↳results_4[randomized_graph_name]
                randomized_x_data =
↳list(randomized_node_properties[x_centrality].values())
                randomized_y_data =
↳list(randomized_node_properties[y_centrality].values())
            else:
                randomized_x_data = []
                randomized_y_data = []

            # Create a scatter plot for original data (blue points)
            axs[row, col].scatter(x_data, y_data, color=COLOR_BLUE,
↳alpha=0.5, label='Original Data')
            axs[row, col].set_xlabel(f"{x_centrality}")
            axs[row, col].set_ylabel(f"{y_centrality}")
            axs[row, col].set_title(f"{x_centrality} vs.
↳{y_centrality}")

            # Create a scatter plot for randomized data (red points)
            axs[row, col].scatter(randomized_x_data, randomized_y_data,
↳color=COLOR_RED, alpha=0.5, label='Randomized Data')

            # Calculate Pearson correlation coefficient for original
↳data
            pearson_corr, _ = pearsonr(x_data, y_data)

```

```

# Calculate Spearman correlation coefficient for original
data spearman_corr, _ = spearmanr(x_data, y_data)

# Calculate Kendall correlation coefficient for original
data kendall_corr, _ = kendalltau(x_data, y_data)

# Calculate Pearson correlation coefficient for randomized
data randomized_pearson_corr, _ = pearsonr(randomized_x_data,
randomized_y_data)

# Calculate Spearman correlation coefficient for randomized
data randomized_spearman_corr, _ = spearmanr(randomized_x_data,
randomized_y_data)

# Calculate Kendall correlation coefficient for randomized
data randomized_kendall_corr, _ = kendalltau(randomized_x_data,
randomized_y_data)

# Generate some descriptive text
corr_text_original = f"Original Data:\nPearson:
{pearson_corr:.4f}\nSpearman: {spearman_corr:.4f}\nKendall: {kendall_corr:.
4f}"

corr_text_randomized = f"Randomized Data:\nPearson:
{randomized_pearson_corr:.4f}\nSpearman: {randomized_spearman_corr:.
4f}\nKendall: {randomized_kendall_corr:.4f}"

axs[row, col].text(0.1, -0.2, corr_text_original,
transform=axs[row, col].transAxes, fontsize=7, ha='center')
axs[row, col].text(0.9, -0.2, corr_text_randomized,
transform=axs[row, col].transAxes, fontsize=7, ha='center')
axs[row, col].legend(loc='upper left') # Add a legend

# Increment the column index and row index when necessary
col += 1
if col >= 3:
    col = 0
    row += 1
else:
    # Skip plotting when the grid is full
    break

```

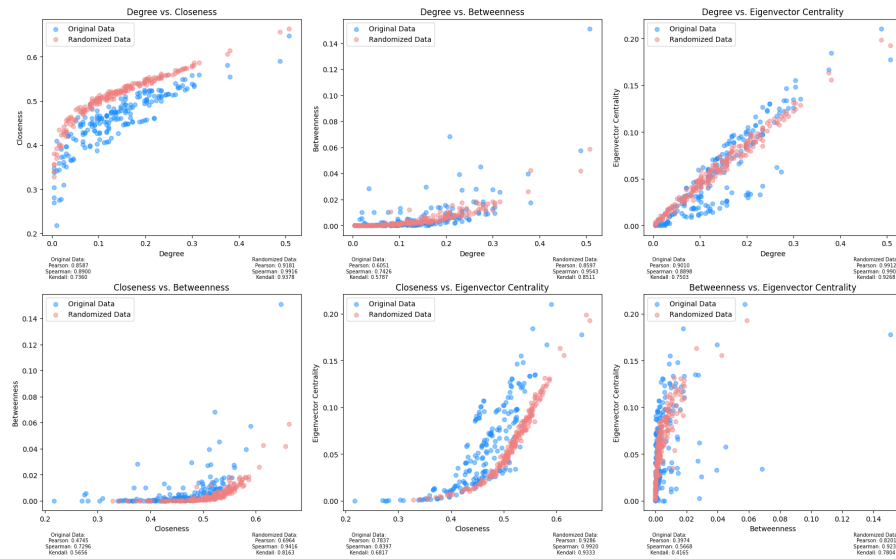
```

# Adjust subplot layout
plt.suptitle(f"Centrality Measures Scatter Plots and Correlation_
Coefficients - {graph_name}", fontsize=16)
plt.tight_layout(rect=[0, 0, 1, 0.95]) # Adjust the supitle position

plt_filename = os.path.join(OUTPUT_PATH,
f"{graph_name}_centralities_super_imposed_explanation.png")
plt.savefig(plt_filename)
plt.show()

```

Centrality Measures Scatter Plots and Correlation Coefficients - graph_jazz_collab



Degree Centrality: - **In practice:** Degree centrality measures the number of collaborations a musician has with other musicians in the Jazz network. - **Meaning:** Musicians with higher degree centrality are those who have collaborated with a larger number of other musicians. They can be seen as more socially active or influential within the Jazz collaboration network.

Closeness Centrality: - **In practice:** Closeness centrality measures how quickly a musician can reach all other musicians in the Jazz network, taking into account the shortest paths. - **Meaning:** Musicians with higher closeness centrality are more central in the network and can quickly interact with a larger portion of other musicians. They may be better positioned to facilitate collaborations and share information across the network.

Betweenness Centrality: - **In practice:** Betweenness centrality identifies musicians who act as “bridges” or intermediaries in the Jazz network, connecting different clusters of musicians. -

Meaning: Musicians with higher betweenness centrality have a significant role in maintaining connections and facilitating communication between groups of musicians who might not otherwise interact. They play a key role in network cohesion.

Eigenvector Centrality: - **In practice:** Eigenvector centrality assigns importance to musicians based on their connections to other influential musicians in the Jazz network. - **Meaning:** Musicians with higher eigenvector centrality are not only well-connected but are also connected to other influential musicians. They are seen as influential themselves and have the potential to impact the network significantly.

```
[241]: # Specify the graph names you want to analyze
graph_names = ["graph_jazz_collab", "graph_jazz_collab_randomized"]

for graph_name in graph_names:
    # Get the centrality measures for the selected graph
    centrality_measures = results_4.get(graph_name, None)

    if centrality_measures is None:
        print(f"Centrality measures not found for {graph_name}")
        continue

    print(f"Centrality Measures for {graph_name}:")

    for measure_name, centrality_values in centrality_measures.items():
        print(f"Centrality Measure: {measure_name}")
        if measure_name == "Degree":
            description = "Degree centrality measures the number of edges_
        connected to a node. Nodes with higher degree centrality are more connected_
        to other nodes."
        elif measure_name == "Closeness":
            description = "Closeness centrality measures how close a node is to_
        all other nodes in the graph. Nodes with higher closeness centrality are_
        more central."
        elif measure_name == "Betweenness":
            description = "Betweenness centrality measures the extent to which_
        a node lies on the shortest paths between other nodes. Nodes with higher_
        betweenness centrality act as bridges in the network."
        elif measure_name == "Eigenvector Centrality":
            description = "Eigenvector centrality measures a node's importance_
        based on the concept that connections to high-scoring nodes contribute more_
        to a node's score. Nodes with higher eigenvector centrality are connected to_
        other influential nodes."
        else:
            description = "Unknown centrality measure"

        centrality_values_rounded = {k: round(v, 4) for k, v in_
        centrality_values.items()}
```

```
print(f"Description: {description}")
# print(f"Centrality Values: {centrality_values_rounded}")
print("\n")
```

Centrality Measures for graph_jazz_collab:

Centrality Measure: Degree

Description: Degree centrality measures the number of edges connected to a node. Nodes with higher degree centrality are more connected to other nodes.

Centrality Measure: Closeness

Description: Closeness centrality measures how close a node is to all other nodes in the graph. Nodes with higher closeness centrality are more central.

Centrality Measure: Betweenness

Description: Betweenness centrality measures the extent to which a node lies on the shortest paths between other nodes. Nodes with higher betweenness centrality act as bridges in the network.

Centrality Measure: Eigenvector Centrality

Description: Eigenvector centrality measures a node's importance based on the concept that connections to high-scoring nodes contribute more to a node's score. Nodes with higher eigenvector centrality are connected to other influential nodes.

Centrality Measures for graph_jazz_collab_randomized:

Centrality Measure: Degree

Description: Degree centrality measures the number of edges connected to a node. Nodes with higher degree centrality are more connected to other nodes.

Centrality Measure: Closeness

Description: Closeness centrality measures how close a node is to all other nodes in the graph. Nodes with higher closeness centrality are more central.

Centrality Measure: Betweenness

Description: Betweenness centrality measures the extent to which a node lies on the shortest paths between other nodes. Nodes with higher betweenness centrality act as bridges in the network.

Centrality Measure: Eigenvector Centrality

Description: Eigenvector centrality measures a node's importance based on the concept that connections to high-scoring nodes contribute more to a node's

score. Nodes with higher eigenvector centrality are connected to other influential nodes.

```
[242]: # !jupyter nbconvert Assignment_01.ipynb --to pdf --output Assignment_01_Results.pdf
```